DATA BASES – PROF. D.M. BRAGA – A. A. 2020/21

3RD SEPTEMBER 2021 – DURATION: 1h 30m

A. SQL (22 points)

The schema below describes the rentals as managed by a Real Estate agency. Ongoing rentals have the EndDate se to NULL until the contract is terminated. The apartment size is expressed in square meters.

CUSTOMER (CusID, LastName, FirstName, BithDate)
APARTMENT (AptID, Street, City, Number, NrOfRooms, AptSize, FloorNr)
RENTAL (AptID, RentalStartDate, RentalEndDate, CusID, PricePerMonth)

1. Specify the command to create table RENTAL, defining the tuple and domain constraints deemed appropriate and expressing all referential integrity constraints towards the other tables (2 p.)
2. Describe briefly but precisely what is extracted by the following query: (4 p.)

```
select City, avg(PricePerMonth/AptSize)
from RENTAL natural join APARTMENT
where RentalStartDate > "3/9/2016"
group by City, Street
having count( distinct AptID ) > 3 and
       avg(PricePerMonth/AptSize) >= ALL ( select avg(PricePerMonth/AptSize)
                                           from RENTAL natural join APARTMENT
                                           where RentalStartDate > "3/9/2016"
                                           group by City, Street
                                           having count( distinct AptID ) > 3
```

3. Extract the customers who returned to an apartment that they had already rented previously. (6pt)
4. Extract the apartments that were never rented by the same customer for more than one year. *For simplicity and brevity, assume that a difference between dates returns an integer that represents the number of elapsed days* (6pt)
5. Extract the apartment that shows the largest percentage increase in renting price from its first to its last rental. *Note: It may be convenient to "stratify" the calculation with the help of one or two views!* (4pt)

B. Conceptual Design (10 points)

- *The ideal schema is complete, legible, minimal, self-explanatory, and of course also correct.*
- *Remember to specify **identifiers** for all entities and **cardinalities** for all associations.*

Civil protection tracks the fires all over Italy. Each fire is assigned an identifier, a starting time, a presumed starting place, and a hazard class. The development of each fire is monitored with variable frequency (its evolution in subsequent instants), characterized by the extension (in hectares), the names of the municipalities involved or endangered, and the current updated hazard class. Each evolution of a fire is also characterized by the current climate conditions (temperature, wind speed and direction) and possibly by containment interventions, carried out by the fire brigade or by air. Interventions by the firefighters cover a period of the evolution of the fire, and are characterized by a commander and the number of participants. Air interventions start from (and return to) an airport located in the territory, have a known duration, and can reach one or more fires (in specific periods of their evolution). At the end of each fire, damages are assessed: hectares of burned land, number of urban settlements destroyed, and possibly injured and dead people.

La protezione civile traccia gli incendi che si sviluppano nelle regioni italiane. Ad ogni incendio è assegnato un identificatore, un istante iniziale, un luogo ove si ipotizza sia iniziato, e una classe di pericolosità. Con periodicità variabile se ne segue lo sviluppo (evoluzione nei successivi istanti di tempo), caratterizzato dall'estensione (in ettari), i nomi dei comuni coinvolti o messi a rischio, e la classe di pericolosità. Ogni evoluzione di un incendio è anche caratterizzata dalle condizioni del clima (temperatura, intensità e direzione del vento) e dagli eventuali interventi di contenimento, effettuati dai vigili del fuoco o per via aerea. Gli interventi dei pompieri coprono un periodo dell'evoluzione dell'incendio, e sono caratterizzati da un comandante e dal numero di partecipanti. Gli interventi aerei partono da (e tornano in) un aeroporto localizzato nel territorio, hanno una durata nota, e possono raggiungere uno o più incendi (in specifici periodi della loro evoluzione). Al termine di ogni incendio si valutano i danni: ettari di terreno coinvolti, numero di insediamenti urbani andati distrutti, eventuali feriti e morti.